Staphylococcus aureus: from raw data to functional annotation

Rana Muhammad Saad[1], Dua Fatima[2], Naima khan[3]

Introduction:

Staphylococcus aureus was selected as the organism for this study because it is an important opportunistic bacterial pathogen with major clinical and public-health significance. It is capable of causing a wide range of infections, from relatively localized skin and soft-tissue infections to severe diseases such as bacteremia, pneumonia, endocarditis, and sepsis. A major concern associated with *S. aureus* is its remarkable ability to acquire and accumulate antimicrobial resistance (AMR) determinants, resulting in multidrug-resistant strains that can substantially limit therapeutic options. In addition to antimicrobial resistance, *S. aureus* possesses numerous virulence factors, including toxins, immune-evasion mechanisms, and biofilm-associated genes, which contribute to its pathogenicity and persistence within the host. Its well-characterized genome, availability of a large number of publicly accessible genome sequences, and extensive representation in genomic databases also make it highly suitable for comparative genomic and pangenomic analyses. Therefore, studying *S. aureus* provides an effective model for investigating the genetic basis of antimicrobial resistance, virulence, genomic diversity, and the potential dissemination of resistance determinants among clinically important bacterial populations.

Dataset Acquisition & Validation

Dataset acquisition and validation was performed to establish a quality-controlled, geographically diverse genome cohort for the *Staphylococcus aureus* capstone project. The process incorporated programmatic genome retrieval from BV BRC Datasets using Python, upload to a shared Galaxy Australia history, and assembly quality assessment using QUAST (Quality Assessment Tool for Genome Assemblies). The project protocol specifies a minimum final cohort of 50 genomes. A total of 60 complete genome assemblies were retrieved and validated, exceeding this minimum by 10 isolates.

Genome Retrieval and Cohort Composition

Genome retrieval was performed programmatically using Python rather than manual selection, allowing systematic, filtered, batch acquisition of assemblies directly from BV BRC Datasets. Retrieval was restricted to isolates with assembly level "Complete Genome" (a fully closed, gap-free genome sequence, as opposed to a fragmented "draft" or "scaffold" assembly), reducing the likelihood of low-quality or heavily fragmented assemblies entering the cohort prior to formal validation.

Geographic diversity was deliberately prioritized during retrieval. Isolates were drawn from multiple

countries and regions, including Pakistan, rather than from a single healthcare system or population. This is significant because *S. aureus* lineage distribution, resistance gene carriage, and plasmid content are known to vary meaningfully by geography; a geographically narrow cohort risks producing a resistome or virulome profile that is not representative of the species as a whole.

A total of 60 complete genome assemblies were retrieved, exceeding the protocol-specified minimum of 50 by 10 isolates. This surplus was retained deliberately rather than narrowed, to provide a larger and more statistically robust dataset for the comparative analyses conducted in subsequent phases (clinical typology, resistome profiling, plasmid co-localization, and pangenome analysis).

Assembly Quality Assessment (QUAST)

All 60 retrieved assemblies were uploaded to a shared Galaxy Australia history, accessible to the full project team, and assessed in batch using QUAST prior to any downstream biological analysis. QUAST evaluation demonstrated consistently high assembly quality across the cohort, confirming the effectiveness of restricting retrieval to "Complete Genome" status. Key summary statistics were as follows:

N50 (a measure of assembly contiguity, where higher values indicate fewer, longer continuous sequence fragments) ranged from 2,735,892 to 3,052,055 bp, with a cohort average of approximately 5,369,597 bp.

Total assembly length ranged from 2,771,008 to 3,052,055 bp, with a cohort average of approximately 5,654,610 bp, consistent with the expected *S. aureus* genome size of 2.7 –3.1 Mb.

Number of contigs per assembly ranged from 1 to 5 contigs, with a cohort average of approximately 3.9, reflecting the near-complete, minimally fragmented nature of the assemblies.

GC content ranged from 32.69 – 33.03% with a cohort average of approximately 32.86%, consistent with established *S. aureus* genomic composition.

No assembly required exclusion on quality grounds. All 60 genomes met a consistently high standard of contiguity and completeness, and the full cohort was therefore retained.

Cohort Reconciliation Note

The protocol specifies a final analytical cohort of 50 genomes. The validated dataset produced at this stage comprises 60 genomes, retained in full because all isolates passed QUAST quality thresholds and no basis for exclusion was identified. This is consistent with the composition reported for the Phase B (Clinical Typology) dataset below, which was supplied with 59 isolates rather than exactly 50. The discrepancy between the protocol minimum and the retained cohort size should be reconciled prior to final submission, either by confirming that the full cohort will be carried through all downstream phases, or by defining a consistent selection criterion for narrowing to exactly 50 isolates across all roles.

Integrated Dataset Summary

Taken together, the genome retrieval and QUAST validation process established a quality-controlled, geographically diverse foundation of 60 complete *S. aureus* genome assemblies, exceeding the project's 50-genome minimum. Retrieval was performed reproducibly via a Python-based workflow with deliberate geographic diversification, including representation from Pakistan, and all assemblies were confirmed to meet high contiguity and completeness standards via QUAST, with no exclusions required. This validated dataset provides the foundation for the clinical-typology, antimicrobial-resistance, plasmid, and whole-genome comparative analyses performed by the remaining project roles.

Clinical Typology: MLST and Virulence Factor Analysis

Multilocus Sequence Typing (MLST)

Methodology

Multilocus sequence typing (MLST) was performed to determine the sequence types (STs) of the selected Staphylococcus aureus isolates. The project cohort consisted of 60 QUAST-selected genome assemblies. For isolates whose ST information was already available, the existing metadata were retained. Five isolates with unknown sequence types were analyzed separately to determine their STs.

Initially, MLST analysis was attempted using the MLST tool available on Galaxy Australia, which supports Staphylococcus aureus. However, the analysis did not successfully provide the required ST results for the unknown isolates. Therefore, the MLST analysis was subsequently performed using the CGE/DTU Denmark platform to determine the sequence types of the remaining isolates.

Challenge Encountered During MLST Analysis

One of the main challenges encountered during this analysis was obtaining the sequence types of the five isolates with previously unknown STs. Although Galaxy Australia provides an MLST tool that supports Staphylococcus aureus, the required results could not be obtained successfully through the initial analysis. As a result, an alternative approach was used, and the genome assemblies were analyzed using the CGE/DTU Denmark MLST platform.

This challenge was resolved by using the alternative MLST platform, which successfully provided the missing sequence-type information.

Table 1. MLST results for isolates with previously unknown ST

Sequence Type (ST)	
SA33-SX	**ST59**
SA01-SX	**ST59**
SA25-SX	**ST59**
SA40TW	**ST59**
NRS70_scf7180000000004	**ST5**

Of the five isolates analyzed, four were assigned to ST59, while **one isolate**, NRS70_scf7180000000004, was assigned to **ST5**.

These results were combined with the previously available ST information for the remaining isolates to contribute to the complete MLST profile of the study cohort.

Virulence Factor Screening Using ABRicate and VFDB

Methodology

Virulence factor screening was performed using **ABRicate** against the Virulence Factor Database (VFDB). The genome assemblies from the selected Staphylococcus aureus cohort were used as input for screening. The analysis was performed to identify virulence-associated genes present within the isolates. The project instructions specifically require Role 2 to screen the genome assemblies using ABRicate with VFDB, or alternatively DTU CGE's Virulence Finder, as part of the virulence profiling workflow.

Summary of ABRicate/VFDB Analysis			
Analysis	**Tool**	**Database**	**Input**
Virulence factor screening assemblies	ABRicate	VFDB	Selected S. aureus genome

Antimicrobial Resistance (AMR) Gene Profiling

Objective

The objective of this analysis was to characterize the antimicrobial resistance (AMR) gene repertoire detected in the supplied Staphylococcus aureus cohort using the StarAMR output generated in Galaxy. The analysis focuses on acquired resistance determinants, their associated predicted antibiotic phenotypes, frequency across isolates, and the overall multidrug-resistance profile.

Data Source and Analysis Approach

The supplied Galaxy file is a StarAMR detailed summary table. It contains records classified as Resistance, Plasmid, and MLST. For this report, only rows with Data Type = Resistance were used for AMR-gene profiling. The resistance records contain isolate ID, resistance determinant, predicted phenotype, sequence identity, overlap, contig, genomic coordinates, and accession information.

- Total isolates represented in the Galaxy output: 60
- Resistance records analyzed: 281
- Unique AMR determinants detected: 25
- Unique isolate–gene associations after removing repeated hits: 231
- Resistance determinants with at least one isolate: 25

The reference report describes StarAMR as a structural and resistance-screening component and specifies a Staphylococcus aureus baseline. The present report does not add thresholds or mutation calls that are

not explicitly represented in the supplied Galaxy table.

Overall AMR Profile

The Galaxy output identified 25 distinct resistance determinants across 60 isolates. The most frequently detected determinant was mecA, found in 48 isolates, followed by blaZ (33 isolates). Multiple macrolide, aminoglycoside, tetracycline, chloramphenicol, antifolate, and fusidic-acid-associated determinants were also present. The distribution indicates that resistance was not restricted to a single antibiotic class.

AMR Gene Frequency and Predicted Phenotype

AMR determinant	Isolates detected	Total hits	Predicted phenotype
mecA	48	48	penicillin
blaZ	33	45	ampicillin
ant(9)-Ia	17	35	spectinomycin
erm(A)	17	33	erythromycin, azithromycin
aph(3')-III	16	16	kanamycin, amikacin
ant(6)-Ia	16	16	streptomycin
erm(B)	14	14	erythromycin, azithromycin
aac(6')-aph(2")	10	11	amikacin, gentamicin, tobramycin
cat(pC221)	8	8	chloramphenicol
tet(K)	8	8	tetracycline
aadD	7	8	amikacin, tobramycin
bleO	7	7	unknown[bleO_1_AF051917]
erm(C)	6	7	erythromycin, azithromycin
dfrG	3	4	trimethoprim
tet(M)	3	3	tetracycline
aph(2")-Ia	2	2	amikacin, gentamicin, tobramycin
cat(pC233)	2	2	chloramphenicol
msr(A)	2	2	erythromycin, azithromycin
fusB	2	2	unknown[fusB_1_AY373761]
dfrE	2	2	unknown[dfrE_1_CP080326]
mph(C)	2	2	erythromycin
cfr	1	1	chloramphenicol, florfenicol, linezolid, clindamycin
lnu(A)	1	1	lincomycin

fusC	1	1	unknown[fusC_1_KF527883]
tet(S/M)	1	1	tetracycline

Major Resistance Categories

Resistance category	Representative determinants	No. of isolates affected
β-lactam / penicillin	mecA, blaZ	56
Macrolide / lincosamide	erm(A), erm(B), erm(C), msr(A), mph(C), lnu(A)	37
Tetracycline	tet(K), tet(M), tet(S/M)	11
Aminoglycoside / related	ant(9)-Ia, aph(3')-III, ant(6)-Ia, aac(6')-aph(2''), aadD, aph(2'')-Ia	38
Phenicols / oxazolidinone	cat(pC221), cat(pC233), cfr	11
Antifolate	dfrG	3

Highest AMR Gene Burden Isolates

Isolate-level burden was calculated as the number of distinct resistance determinants detected per isolate. Repeated hits of the same determinant within an isolate were counted once for this purpose.

Isolate ID	Distinct AMR determinants	Detected determinants
SR153	10	aac(6')-aph(2''), aadD, ant(9)-Ia, blaZ, bleO, cfr, erm(A), lnu(A), mecA, tet(K)
XN108	9	aac(6')-aph(2''), ant(6)-Ia, ant(9)-Ia, aph(3')-III, blaZ, dfrG, erm(A), mecA, tet(M)
VB9352	8	aac(6')-aph(2''), ant(6)-Ia, aph(3')-III, blaZ, dfrG, mecA, mph(C), msr(A)
NY2010	7	aac(6')-aph(2''), ant(9)-Ia, blaZ, erm(A), erm(C), mecA, tet(M)
SR130	7	ant(6)-Ia, aph(2'')-Ia, aph(3')-III, blaZ, dfrE, erm(B), mecA
NY2491	7	aac(6')-aph(2''), ant(9)-Ia, blaZ, erm(A), erm(C), mecA, tet(M)

R46	7	aac(6')-aph(2''), aadD, blaZ, bleO, dfrG, erm(C), mecA
ZY05	7	ant(6)-Ia, aph(3')-III, blaZ, cat(pC233), erm(B), mecA, tet(K)
SA0907	6	aac(6')-aph(2''), blaZ, erm(C), fusC, mecA, tet(K)
P2D15C1	6	aadD, ant(9)-Ia, blaZ, bleO, erm(A), mecA

Interpretation of Key AMR Determinants

mecA: Detected in 48 isolates and associated in the Galaxy output with a penicillin phenotype. It is the most prevalent resistance determinant in this dataset.

blaZ: Detected in 33 isolates as a unique-isolate count, with 45 total hits. The output associates it with an ampicillin phenotype; repeated hits occur in some isolates.

erm(A), erm(B), erm(C): These macrolide-associated determinants were detected in 17, 14, and 6 isolates respectively and were linked in the output to erythromycin/azithromycin phenotypes.

ant(9)-Ia: Detected in 17 isolates and associated with spectinomycin in the Galaxy phenotype field.

ant(6)-Ia: Detected in 16 isolates and associated with streptomycin.

aph(3')-III: Detected in 16 isolates and associated with kanamycin/amikacin.

aac(6')-aph(2''): Detected in 10 isolates and associated with amikacin/gentamicin/tobramycin.

tet(K), tet(M), tet(S/M): Tetracycline-associated determinants detected in 8, 3, and 1 isolates, respectively.

dfrG: Detected in 3 isolates and associated with trimethoprim.

cfr: Detected in 1 isolate and associated in the output with chloramphenicol, florfenicol, linezolid, and clindamycin.

Sequence Match Quality

The resistance records show generally high sequence identity and overlap. However, the Galaxy table contains some hits with identity below 100% and, for selected determinants such as blaZ, erm(C), and ant(6)-Ia, overlap values below 100% are present. Therefore, the statement that every detected determinant has 100% identity and overlap would not accurately describe this Galaxy output. Observed resistance-hit identity range: 98.46%–100.00%. Observed overlap range: 63.12%–100.23%.

Chromosomal Point Mutations

No separate chromosomal point-mutation records are represented in the supplied StarAMR detailed summary under Data Type = Resistance. Consequently, this report does not assign specific point mutations (for example, grlA substitutions) to the isolates. Such mutations should only be reported if they are present

in the corresponding StarAMR mutation/PointFinder output.

Conclusion

The supplied Galaxy StarAMR dataset demonstrates a broad AMR profile involving 25 distinct resistance determinants across 60 isolates. mecA was the most widely distributed determinant, while blaZ and several macrolide- and aminoglycoside-associated genes were also common. Tetracycline and antifolate resistance determinants were detected in smaller subsets, and a limited number of determinants were associated with phenotypes involving chloramphenicol, florfenicol, linezolid, clindamycin, or fusidic-acid-associated/unknown labels.

Overall, the dataset supports the presence of multidrug-resistance determinants in the cohort. The results should be interpreted as genomic resistance predictions from the StarAMR output rather than direct antimicrobial susceptibility testing. For a complete resistome assessment, the AMR-gene results can be integrated with the plasmid records and, where available, Point Finder mutation results to evaluate potential mobility and chromosomal resistance mechanisms.

PLASMID & CO-LOCALIZATION ANALYSIS

Methods

PlasmidFinder Analysis

We identified plasmid replicon types across all 60 *Staphylococcus aureus* genomes using PlasmidFinder (via Galaxy). We applied the tool to the full genome collection to detect replicon sequences indicative of plasmid presence. Replicon type and frequency were recorded per genome.

AMR–Plasmid Co-localization

Antimicrobial resistance (AMR) gene locations were determined using ABRicate and StarAMR on the same 60-genome collection. AMR-bearing contigs were cross-referenced against PlasmidFinder replicon-bearing contigs. A co-localization event was defined as the occurrence of an AMR gene and a PlasmidFinder replicon on the same contig within the same genome. Results were compiled into a structured co-localization table (AMR_plasmid_colocalization.csv) containing genome ID, plasmid replicon type, plasmid contig accession, AMR gene name, AMR contig accession, and co-localization status.

VF–Plasmid Co-localization

Virulence factor (VF) gene locations were identified using a dedicated VF database screen. VF-bearing contigs were compared to PlasmidFinder replicon-bearing contigs using the same contig-matching approach. Results were compiled into VF_plasmid_colocalization.csv. For both AMR and VF analyses, the distinction between chromosome-scale contigs and small plasmid-scale contigs was assessed by

examining contig accession identifiers and comparing contig identity between the VF/AMR contig and the plasmid-replicon contig columns.

Results

PlasmidFinder Replicon Distribution

Of the 60 genomes analysed, **41 harboured at least one identifiable PlasmidFinder replicon**; the remaining **19 genomes yielded no detected replicon**. A total of **14 distinct replicon types** were identified. The most prevalent replicons in the AMR-associated subset were rep22 (n = 49), rep16 (n = 37), rep7a (n = 30), rep7c (n = 24), and rep19 (n = 24) (Table 1).

Table 1. PlasmidFinder replicon types and frequencies in AMR co-localization instances

Plasmid Replicon	AMR Co-localization Instances
rep22	49
rep16	37
rep7a	30
rep7c	24
rep19	24
rep5a	14
rep21	14
rep20	14
rep10	11
rep7b	10
repUS43	6
Col156	5
repUS22	4
rep15	2
Total	244

AMR–Plasmid Co-localization

The validated AMR co-localization dataset contained **244 co-localization instances** spanning **24 distinct AMR genes** across **37 genomes**. All 244 instances were confirmed co-localized. **tet(M)** was absent from the dataset entirely — it had no plasmid-contig association in this collection.

The most frequently plasmid-associated AMR genes were *blaZ* (n = 67), *mecA* (n = 22), *tet(K)* (n = 20), *ant(9)-Ia* (n = 19), and *erm(A)* (n = 19) (Table 2).

Table 2. AMR genes and plasmid co-localization frequencies

AMR Gene	Instances	Key Replicons
blaZ	67	rep16, rep19, rep5a, rep7c
mecA	22	rep22, rep7c, rep21
tet(K)	20	rep7a, rep16, rep19
ant(9)-Ia	19	rep22, rep7a, rep7c
erm(A)	19	rep22, rep7a, rep7c
aac(6')-aph(2'')	13	rep22, rep21, rep7a
aadD	11	rep22, rep7a
erm(C)	11	rep22
bleO	8	rep22, rep7a
cat(pC221)	7	rep16, rep19
fusB	6	rep16, rep21, rep7c
erm(B)	6	rep20, rep16
aph(2'')-Ia	4	rep16, rep19
dfrE	4	rep16, rep5a
ant(6)-Ia	4	rep22, rep7b
aph(3')-III	4	rep22, rep7b
lnu(A)	4	rep22, rep7a

AMR Gene	Instances	Key Replicons
tet(S/M)	3	rep7a
fusC	3	rep21, rep7a, rep20
dfrG	2	Col156, rep7c
cat(pC233)	2	rep7b
mph(C)	2	rep16, rep19
msr(A)	2	rep16, rep19
cfr	1	rep22
tet(M)	0 (absent)	None identified
Total	244	

Several contigs harboured multiple AMR genes co-localizing with the same replicon, indicating multi-resistance plasmid-scale elements. Notable clusters include:

• **NRS70_scf7180000000004 / P2D15C1 / P2D1C1 / SR153 / ZJ5499** (rep22): *aadD*, *ant(9)-Ia*, *bleO*, *erm(A)*, *mecA* — five AMR genes on a conserved rep22 replicon-bearing contig across five independent genomes.

• **R46** (CP039164.1, Col156/rep7c): *aac(6')-aph(2'')*, *blaZ*, *bleO*, *dfrG*, *mecA*.

• **SA0907** (CP121204.1, rep20/rep21/rep7a): *aac(6')-aph(2'')*, *blaZ*, *fusC*, *mecA*, *tet(K)*.

• **USA300_2014.C01** (NZ_CP012118.1, rep16/rep19): *ant(6)-Ia*, *aph(3')-III*, *blaZ*, *mph(C)*, *msr(A)*.

• **SA03-SX** (CP130549.1, rep7a/rep7c/repUS43): *ant(9)-Ia*, *erm(A)*, *tet(K)*, *tet(S/M)*.

VF–Plasmid Co-localization

The VF co-localization dataset contained **2,705 contig-level match instances** involving **96 unique VF genes** across **20 genomes**. All 2,705 instances were recorded as co-localized.

Critical interpretation note: In 100% of cases (2,705/2,705), the VF contig accession was identical to the plasmid replicon contig accession. These contig identifiers correspond to chromosome-scale assemblies (e.g., CP130549.1, CP047321.1, CP082815.1). **No VF gene was found on a distinct small plasmid contig.** The PlasmidFinder replicon signals and VF genes co-occur on the same large chromosomal contigs — not on discrete plasmid elements.

Table 3. Summary of VF–plasmid co-localization findings

Parameter	Value
Total co-localization instances	2,705
Unique VF genes with genome+contig match	96
Genomes with VF co-localization matches	20 / 60
VF genes on distinct small plasmid contigs	0
Nature of all VF–replicon contig matches	Chromosome-scale contigs only
Evidence of VF genes being plasmid-borne	None

Discussion

AMR Co-localization and Plasmid-Mediated Resistance

The finding that all 24 AMR genes in the dataset had at least one confirmed plasmid replicon co-localization is consistent with the well-established role of plasmids in horizontal AMR dissemination in *S. aureus*. The rep22-associated five-gene resistance cluster (*aadD*, *ant(9)-Ia*, *bleO*, *erm(A)*, *mecA*) detected across five independent genomes is particularly notable, suggesting a conserved multi-resistance plasmid that has spread clonally or via horizontal gene transfer.

The high frequency of *blaZ* (n = 67) associated with rep16, rep19, and rep5a replicons reflects the known prevalence of beta-lactamase plasmids in *S. aureus*. The association of *mecA* with rep22 and rep7-family replicons is similarly consistent with known SCCmec and plasmid-integrated resistance contexts.

The sole AMR gene without any plasmid co-localization, **tet(M)**, was absent from the co-localization dataset entirely. This may reflect chromosomal integration, an integrative conjugative element not detected by PlasmidFinder, or absence from genomes harbouring identifiable replicons. This warrants further investigation using complementary databases (e.g., ICEfinder, ISfinder).

VF Co-localization: Absence of Plasmid-Borne Evidence

Although 96 VF genes registered genome+contig matches with PlasmidFinder replicons in 20 genomes, this does not constitute evidence of plasmid-borne virulence. The identical contig accession numbers in both the VF and plasmid replicon columns, combined with the chromosome-scale nature of these contigs, indicates that PlasmidFinder detected replicon-homologous sequences within the chromosome

rather than on discrete plasmid elements.

This most likely reflects chromosomal integration of replicon-homologous sequences (known to occur in *S. aureus* pathogenicity islands and SCC elements) or incidental sequence similarity between chromosomal regions and replicon database entries. Under either interpretation, the data **do not support the conclusion that VF genes are plasmid-borne or horizontally transferable via plasmid** in this dataset.

Co-localization vs. Confirmed Plasmid Transferability

Co-localization of AMR genes with PlasmidFinder replicons on the same contig is necessary but not sufficient evidence of plasmid mobility. Confirmed transferability additionally requires evidence of: (a) plasmid circularity or extrachromosomal existence; (b) conjugation or mobilization genes; and (c) observed horizontal transfer in experimental or phylogenetic studies. The co-localization data support the *potential* mobility of AMR genes but should not be interpreted as direct proof of active transfer.

Limitations

1. Contig resolution: Several genomes were represented by draft assemblies (e.g., NZ_JXZM01000001.1), where plasmid and chromosomal sequences may be co-assembled, potentially inflating co-localization counts.

2. PlasmidFinder database coverage: Replicons with novel or highly divergent sequences will be missed. The 19 genomes with no detected replicon may nonetheless carry unidentified plasmids.

3. VF contig scale: No size-filtering of contigs was performed before co-localization analysis, preventing reliable distinction between small plasmid contigs and large chromosomal contigs bearing replicon-homologous sequences.

4. tet(M) interpretation: The absence of tet(M) from the co-localization table reflects dataset-specific absence rather than a universal biological property; this gene is known to be plasmid-associated in other *S. aureus* collections.

5. No functional validation: All findings are computational. No wet-laboratory validation (plasmid curing, conjugation assays, Southern blot) was performed.

Conclusion

Analysis of 60 *S. aureus* genomes identified PlasmidFinder replicons in **41 genomes (68.3%)**, with **14 distinct replicon types** detected. All 24 AMR genes present in the co-localization dataset had confirmed associations with at least one plasmid replicon-bearing contig, supporting the role of plasmids in AMR gene dissemination. The rep22-associated five-gene resistance cluster (*aadD*, *ant(9)-Ia*, *bleO*, *erm(A)*, *mecA*) observed across multiple independent genomes represents a particularly significant multi-

resistance plasmid candidate. *tet(M)* had no plasmid association in this dataset and warrants further investigation.

For virulence factors, while 96 VF genes registered contig-level co-localization matches with plasmid replicons in 20 genomes, **all such matches occurred on chromosome-scale contigs**. No VF gene was detected on a distinct small plasmid-scale contig. Accordingly, this analysis provides **no evidence that VF genes are plasmid-borne** in this *S. aureus* dataset. These findings highlight the importance of contig-scale interpretation in plasmid co-localization studies.

iTOL & PAN-GENOME ANALYSIS REPORT

Number of strains	60
Total gene clusters	7,776
Core / Shell / Soft-core / Cloud	1,700 / 1,343 / 235 / 4,498
Analyses represented	parSNP tree, Roary pangenome, iTOL MLST/country annotation

Introduction

Comparative genome analysis provides a framework for examining genetic relatedness, gene-content diversity, and population structure among bacterial strains. In this analysis, 60 strains were evaluated using a combination of a parSNP phylogenetic tree, Roary-based pangenome analysis, and an iTOL circular visualization annotated with multilocus sequence typing (MLST) sequence types and country information. The supplied figures show 7,776 gene clusters in the Roary matrix and classify these clusters into core, shell, soft-core, and cloud components.

Objectives

- To assess phylogenetic relationships among the 60 bacterial strains.
- To characterize the overall pangenome and quantify core and accessory gene components.
- To visualize strain-associated MLST sequence types and geographic annotations using iTOL.
- To interpret the extent of genomic conservation and gene-content variability within the dataset.

Dataset and Analytical Approach

The supplied results represent 60 strains. The phylogenetic visualization is labeled “parSNP tree (60 strains),” while the Roary matrix is labeled “Roary matrix (7776 gene clusters).” The pangenome plot further divides the 7,776 clusters according to their distribution across strains. The iTOL visualization adds two annotation tracks: MLST sequence type and country.

parSNP phylogenetic analysis

The parSNP tree summarizes relationships among the 60 strains based on shared single-nucleotide polymorphism information. Branching patterns indicate relative genetic similarity: strains joined by

shorter or more closely positioned branches are more closely related within the represented dataset, whereas separated branches indicate greater divergence. The supplied figure shows multiple clusters of closely related strains rather than a single homogeneous lineage.

Roary pangenome analysis

Roary groups homologous genes into gene clusters and describes the pangenome according to how widely each cluster is distributed among the analyzed strains. In the supplied output, 7,776 gene clusters were identified. The distribution plot shows a high number of clusters at the extremes of strain frequency, with many clusters present in only a small number of genomes and a substantial group present in almost all or all genomes.

iTOL visualization

The iTOL figure displays the strain relationships in a circular layout and overlays MLST sequence types and countries as annotation tracks. The legend includes multiple MLST sequence types and countries including Taiwan, Pakistan, Japan, Saudi Arabia, China, Israel, Kazakhstan, Thailand, Georgia, and India. These annotations facilitate visual comparison of lineage and geographic metadata.

Results

Integrated phylogenetic tree and Roary gene-presence matrix

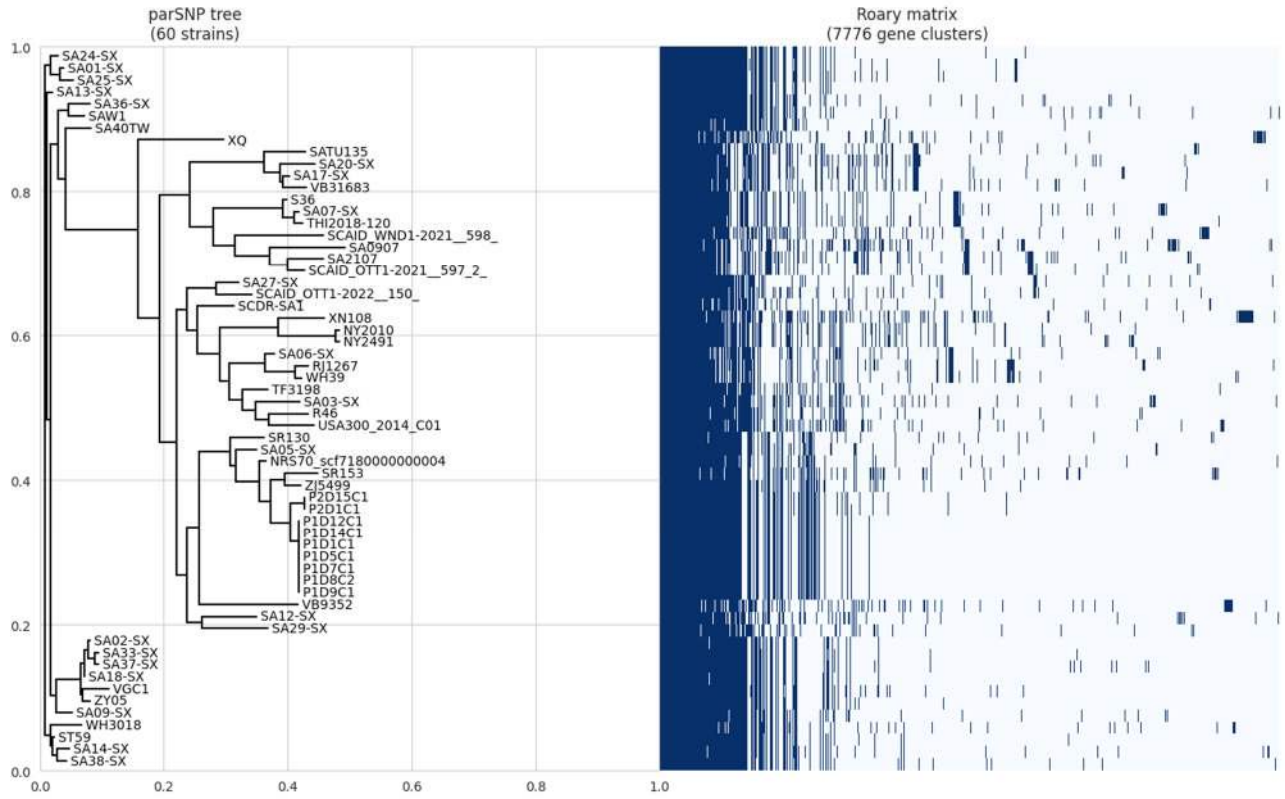


Figure 1. parSNP phylogenetic tree of 60 strains alongside the Roary gene-presence/absence matrix containing 7,776 gene clusters.

Figure 1 integrates the strain-level phylogenetic structure with the corresponding Roary gene-content matrix. The matrix contains dense regions of shared genes as well as numerous sparse patterns, indicating that the strains share a substantial genomic backbone but also possess variable accessory genes. The phylogenetic component displays several distinguishable groups, demonstrating genetic structure within the 60-strain collection.

Distribution of gene clusters across genomes

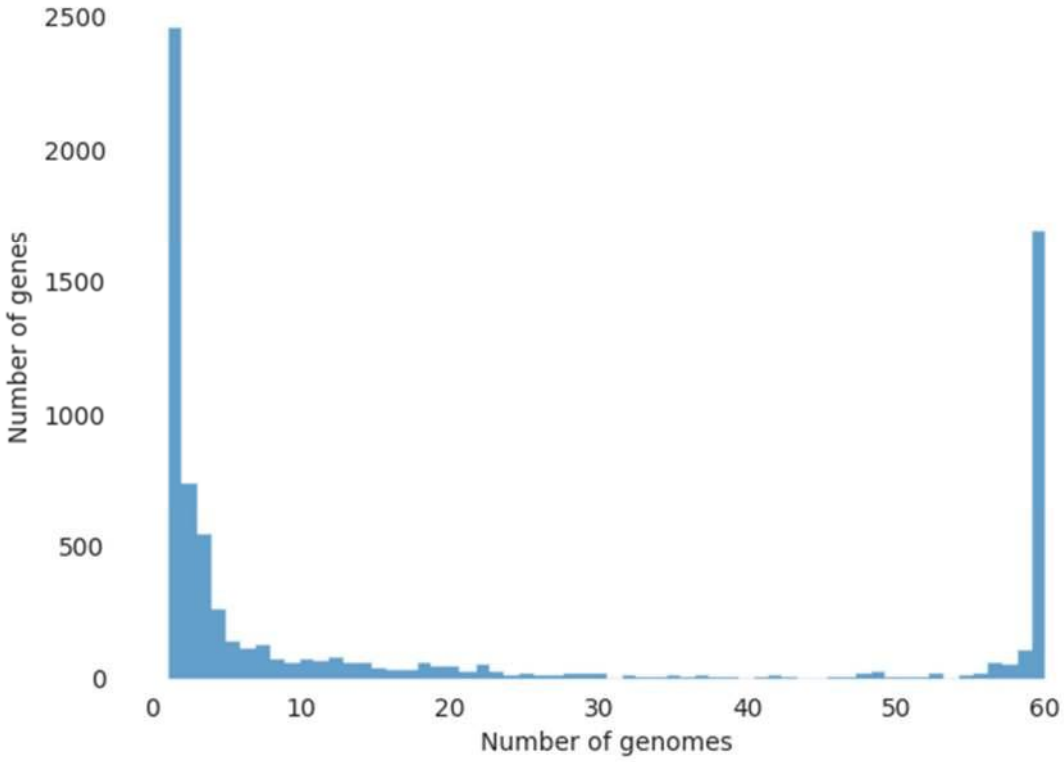


Figure 2. Distribution of Roary gene clusters according to the number of genomes in which each cluster occurs.

The gene-frequency distribution demonstrates that gene clusters are not uniformly distributed among the 60 genomes. A pronounced peak occurs at low genome frequencies, reflecting numerous accessory genes that occur in relatively few strains. A second prominent increase is observed near the high-frequency end, consistent with a large set of genes shared by most genomes. This pattern is characteristic of a pangenome containing both a conserved genomic component and a substantial accessory component.

Pangenome composition

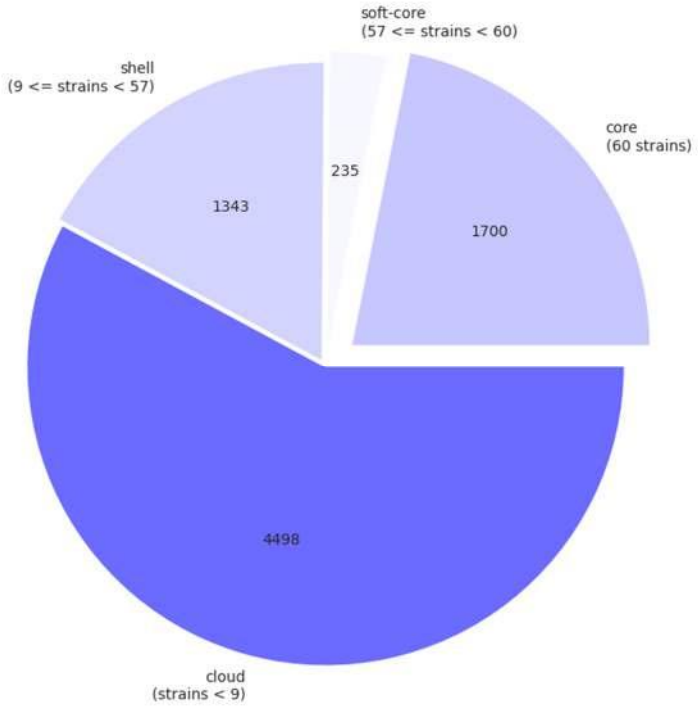


Figure 3. Pangenome composition showing core, soft-core, shell, and cloud gene clusters.

Pangenome category	Gene clusters	Percentage
Core	1,700	21.86%
Soft-core	235	3.02%
Shell	1,343	17.27%
Cloud	4,498	57.84%

The 7,776 gene clusters comprise 1,700 core genes (21.86%), 235 soft-core genes (3.02%), 1,343 shell genes (17.27%), and 4,498 cloud genes (57.84%). Thus, the largest fraction of the pangenome is represented by cloud genes, which occur in fewer than nine strains in the supplied classification. The core and soft-core components together account for 1,935 clusters (24.88%), representing the more broadly conserved portion of the gene repertoire.

iTOL MLST and geographic annotation

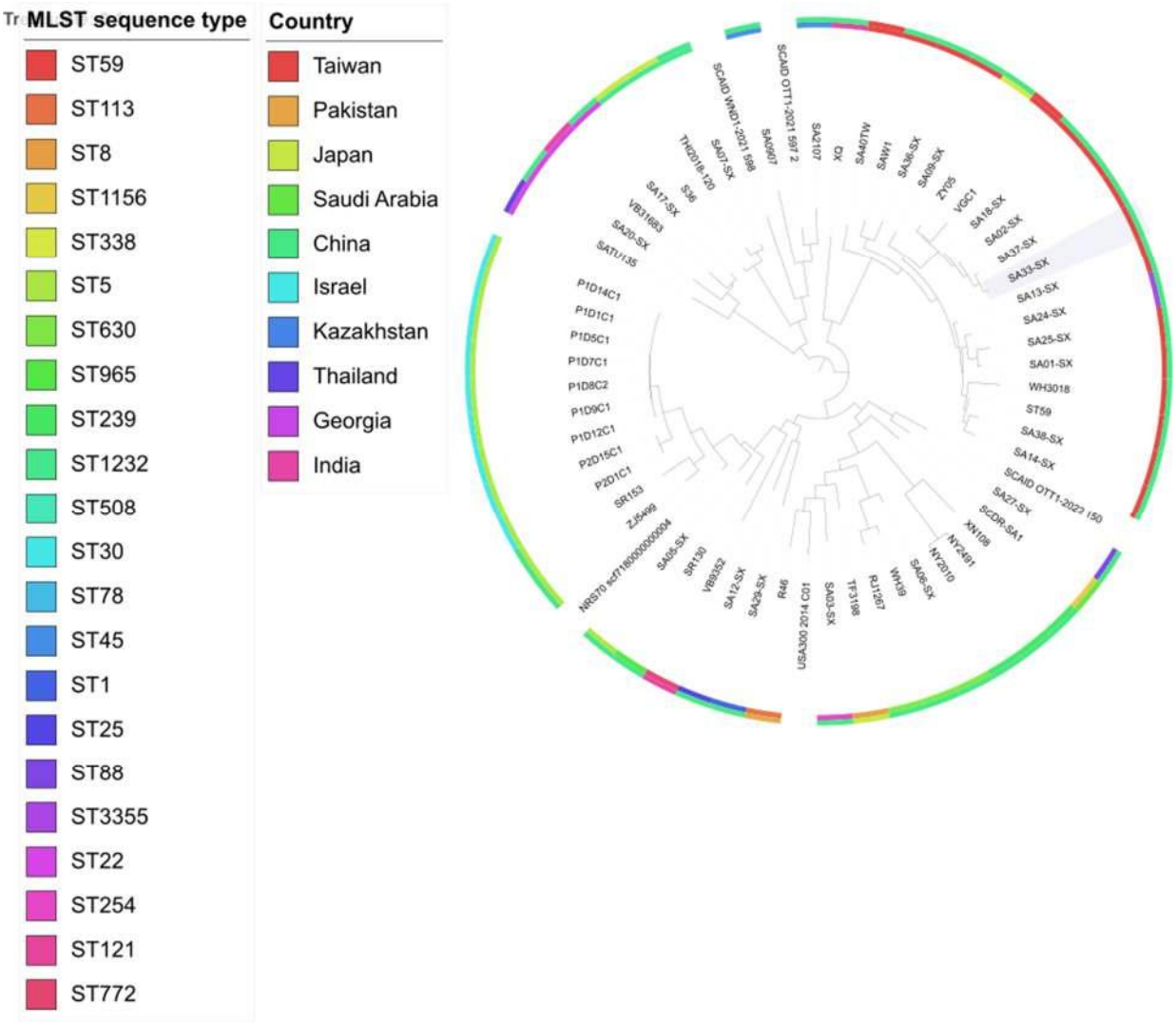


Figure 4. iTOL circular visualization annotated with MLST sequence types and country metadata.

The iTOL visualization provides a complementary population-level view. The MLST legend contains a broad range of sequence types, including ST59, ST113, ST8, ST1156, ST338, ST5, ST630, ST965, ST239, ST1232, ST508, ST30, ST78, ST45, ST1, ST25, ST88, ST3355, ST22, ST254, ST121, and ST772. The country legend includes Taiwan, Pakistan, Japan, Saudi Arabia, China, Israel, Kazakhstan, Thailand, Georgia, and India. The co-occurrence of multiple sequence types and countries indicates substantial population diversity in the analyzed collection.

Discussion

The combined results indicate that the 60-strain collection has considerable genomic diversity. The phylogenetic tree separates the strains into several groups, while the Roary matrix reveals extensive variation in gene presence and absence. Importantly, more than half of all detected gene clusters belong to the cloud category (57.84%), whereas the strictly conserved core comprises 21.86%. This contrast suggests that accessory gene content contributes substantially to differences among strains.

The distribution of gene frequencies supports this interpretation: many clusters occur in only a small

subset of genomes, while another large group is found at very high frequency. Such a distribution indicates a conserved genomic backbone accompanied by a diverse accessory genome. Accessory genes may contribute to strain-specific phenotypes, adaptation, or ecological specialization; however, functional conclusions about individual genes cannot be established from the supplied figures alone. The iTOL annotations further show that the dataset contains multiple MLST sequence types and isolates associated with several countries. Geographic and MLST annotations should be interpreted as metadata rather than proof of geographic transmission or evolutionary origin. Establishing such relationships would require sampling design, temporal information, and additional epidemiological or genomic analyses.

Key Findings

- 60 bacterial strains were included in the comparative analysis.
- The Roary output contains 7,776 gene clusters.
- Core genes: 1,700 clusters (21.86%).
- Soft-core genes: 235 clusters (3.02%).
- Shell genes: 1,343 clusters (17.27%).
- Cloud genes: 4,498 clusters (57.84%), the largest pangenome fraction.
- The parSNP tree displays multiple phylogenetic groupings among the strains.
- The iTOL annotation includes numerous MLST sequence types and isolates from multiple countries.

Conclusion

The integrated iTOL and pangenome analysis demonstrates substantial genomic heterogeneity among the 60 analyzed strains. Although a conserved core genome of 1,700 clusters is present, the pangenome is dominated by accessory cloud genes (4,498 clusters), highlighting extensive gene-content variability. The parSNP phylogeny and iTOL annotations further demonstrate structured genetic diversity across the collection, with multiple MLST sequence types and geographic origins represented. Overall, the results provide a strong comparative-genomic overview and establish a basis for subsequent analyses such as functional annotation of accessory genes, virulence/resistance profiling, or detailed population-genomic investigation.

Functional Annotation and Pathway Reconstruction Report

Executive Summary

Functional annotation was performed to characterize the biological capabilities encoded by the analyzed Staphylococcus aureus genome dataset. Two complementary annotation frameworks were evaluated: Clusters of Orthologous Groups (COG) functional classification and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway reconstruction. The supplied COG output classified the core-gene set across

21 functional categories, with the largest proportion assigned to the category S (function unknown; 21.34%), followed by amino acid transport and metabolism (E; 8.76%), translation/ribosomal structure and biogenesis (J; 8.58%), transcription (K; 7.29%), and inorganic ion transport and metabolism (P; 7.23%).

The KEGG Mapper reconstruction identified 212 pathway entries, together with 36 BRITE categories, 8 BRITE tables, and 37 modules. The strongest pathway-level signals were associated with general metabolism, secondary-metabolite biosynthesis, microbial metabolism in diverse environments, cofactor biosynthesis, amino-acid biosynthesis, carbon metabolism, ribosomal functions, ABC transporters, two-component systems, and peptidoglycan biosynthesis. The reconstruction also mapped functions to Staphylococcus aureus infection and antimicrobial-resistance pathways, including cationic antimicrobial peptide, beta-lactam, vancomycin, and antifolate resistance.

Together, the COG and KEGG profiles indicate a metabolically versatile bacterial genome repertoire supported by extensive information-processing, transport, cell-envelope, stress-response, and regulatory functions. The results should be interpreted as functional potential inferred from gene annotations rather than direct evidence that every reconstructed pathway is fully active under a particular growth condition.

Objective

The objective of this analysis was to establish a comprehensive functional profile of the S. aureus genome dataset by (i) classifying core genes into COG functional categories, (ii) assigning KEGG Orthology-linked functions to pathway maps, modules, and BRITE classifications, and (iii) integrating the two annotation systems to identify major biological capabilities relevant to cellular maintenance, metabolism, environmental adaptation, cell-envelope biology, and antimicrobial resistance.

Methodology

COG Functional Classification

The supplied COG result consisted of percentage-based functional assignments for the core-gene set. Each gene assignment was represented by a standard single-letter COG category. The reported percentages were used directly to describe the functional distribution. Because the supplied result is expressed as percentages, the analysis focuses on relative functional representation rather than absolute gene counts.

KEGG Pathway Reconstruction

The KEGG analysis was performed using KEGG Mapper Reconstruction with the prepared KO input. The resulting HTML report records matched KEGG Orthology identifiers against pathway maps and reports the number of pathway, BRITE, BRITE-table, and module entries containing matched objects. The final reconstruction contained 212 pathway entries, 36 BRITE entries, 8 BRITE tables, and 37 modules.

COG Results

The COG profile demonstrates broad functional coverage across housekeeping, metabolic, regulatory, envelope, and defense-associated processes. The distribution is dominated by category S, indicating that a substantial fraction of the core-gene assignments could not be attributed to a more specific COG functional class. Among the functionally characterized categories, amino acid metabolism, translation, transcription, inorganic ion metabolism, energy production, carbohydrate metabolism, and cell-envelope biogenesis were prominent.

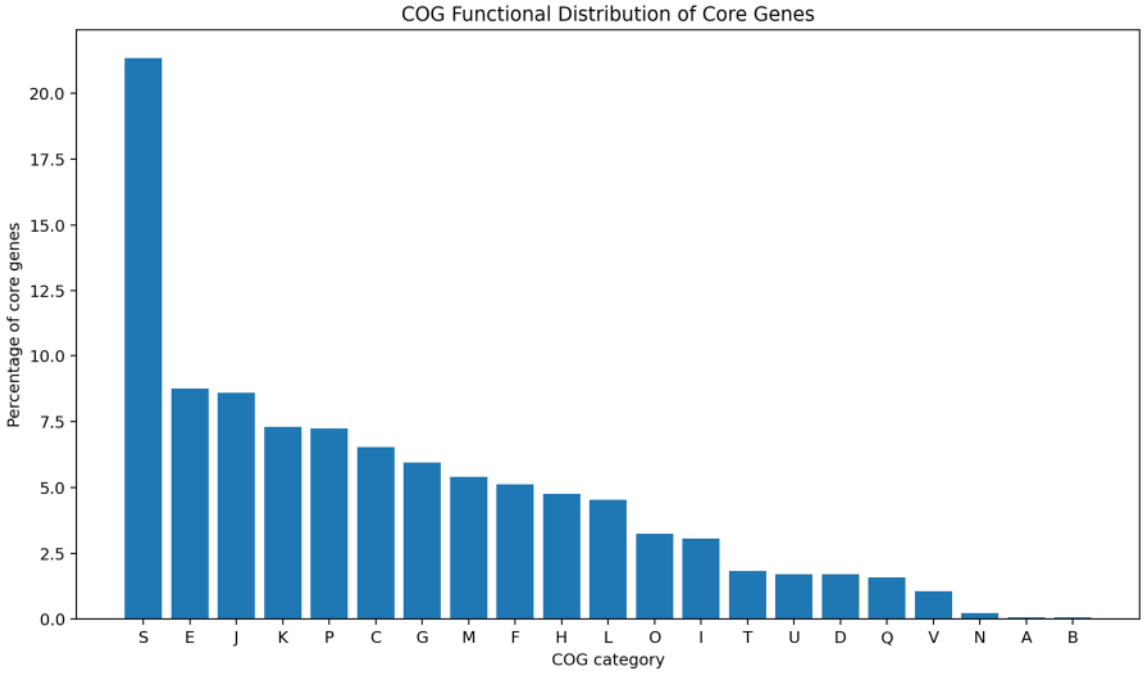


Figure 1. Distribution of COG functional categories among the supplied core-gene results.

Detailed COG Distribution

COG	Functional category	Percentage (%)
S	Function unknown	21.34
E	Amino acid transport and metabolism	8.76
J	Translation, ribosomal structure and biogenesis	8.58
K	Transcription	7.29
P	Inorganic ion transport and metabolism	7.23
C	Energy production and conversion	6.53
G	Carbohydrate transport and metabolism	5.94

M	Cell wall/membrane/envelope biogenesis	5.41
F	Nucleotide transport and metabolism	5.11
H	Coenzyme transport and metabolism	4.76
L	Replication, recombination and repair	4.53
O	Posttranslational modification, protein turnover and chaperones	3.23
I	Lipid transport and metabolism	3.06
T	Signal transduction mechanisms	1.82
U	Intracellular trafficking, secretion and vesicular transport	1.70
D	Cell cycle control, cell division and chromosome partitioning	1.70
Q	Secondary metabolites biosynthesis, transport and catabolism	1.59
V	Defense mechanisms	1.06
N	Cell motility	0.24
A	RNA processing and modification	0.06
B	Chromatin structure and dynamics	0.06

The largest characterized category was E (amino acid transport and metabolism; 8.76%), suggesting substantial capacity for amino-acid acquisition, biosynthesis, and utilization. Category J (8.58%) reflects the importance of translation and ribosomal functions, while K (7.29%) indicates a sizeable transcriptional regulatory and RNA-synthesis component. Category P (7.23%) represents inorganic-ion transport and metabolism, consistent with the importance of ion homeostasis in bacterial physiology. Energy production and conversion (C; 6.53%) and carbohydrate transport and metabolism (G; 5.94%) further indicate strong central metabolic capacity.

Cell wall, membrane, and envelope biogenesis (M; 5.41%) is particularly relevant to S. aureus because the cell envelope is central to structural integrity, host interaction, and susceptibility to several

antimicrobial classes. Nucleotide metabolism (F; 5.11%), coenzyme metabolism (H; 4.76%), and DNA replication/recombination/repair (L; 4.53%) provide additional evidence of broad biosynthetic and genome-maintenance capabilities.

Defense mechanisms (V) accounted for 1.06% of the core-gene assignments. Although this category is relatively small, defense-associated functions can have disproportionate biological importance because individual systems may contribute to resistance against antimicrobial compounds, host-associated stresses, or competing organisms.

KEGG Results

KEGG Mapper reconstructed 212 pathway entries from the supplied KO set. The result also contained 36 BRITE categories, 8 BRITE tables, and 37 modules. The pathway reconstruction was dominated by metabolic maps, followed by genetic-information-processing, environmental-information-processing, cellular-process, organismal-system, and human-disease pathway maps.

Major Metabolic Functions

Metabolism represented the strongest component of the KEGG reconstruction. The broad 'Metabolic pathways' map contained 444 matched KO objects, while 'Biosynthesis of secondary metabolites' contained 219 and 'Microbial metabolism in diverse environments' contained 115. Other prominent maps included biosynthesis of cofactors (95), biosynthesis of amino acids (85), carbon metabolism (64), purine metabolism (42), nucleotide metabolism (33), peptidoglycan biosynthesis (33), and pyruvate metabolism (31).

KEGG map	Pathway	Matched KO objects
01100	Metabolic pathways	444
01110	Biosynthesis of secondary metabolites	219
01120	Microbial metabolism in diverse environments	115
01240	Biosynthesis of cofactors	95
01230	Biosynthesis of amino acids	85
01200	Carbon metabolism	64
00230	Purine metabolism	42
01232	Nucleotide metabolism	33
00550	Peptidoglycan biosynthesis	33
00620	Pyruvate metabolism	31

The presence of peptidoglycan biosynthesis among the strongly represented pathways is biologically important for a Gram-positive organism such as S. aureus. The combination of carbon metabolism,

pyruvate metabolism, amino-acid biosynthesis, nucleotide metabolism, and cofactor biosynthesis indicates broad coverage of pathways needed for biomass formation and cellular maintenance.

Genetic Information Processing

The genetic-information-processing component included a strong ribosomal signal, with 53 matched KO objects in the ribosome pathway. Aminoacyl-tRNA biosynthesis contained 21, homologous recombination 17, protein export 14, mismatch repair 13, and DNA replication 12. These findings are consistent with an extensive repertoire for protein synthesis, genome maintenance, DNA repair, and protein translocation.

KEGG map	Pathway	Matched KO objects
03010	Ribosome	53
00970	Aminoacyl-tRNA biosynthesis	21
03440	Homologous recombination	17
03060	Protein export	14
03430	Mismatch repair	13

Environmental Information Processing

Environmental-information-processing pathways were represented particularly strongly by ABC transporters (57 matched KO objects) and the two-component system (56). The phosphotransferase system (PTS) contained 21 matched objects, while bacterial secretion systems contained 12. These systems are relevant to nutrient uptake, signal sensing, adaptation to environmental conditions, and transport of molecules across the bacterial envelope.

KEGG map	Pathway	Matched KO objects
02010	ABC transporters	57
02020	Two-component system	56
02060	Phosphotransferase system (PTS)	21
03070	Bacterial secretion system	12

Cellular Processes and Community-Associated Functions

Quorum sensing was represented by 29 matched KO objects, making it one of the most prominent cellular-process maps in the reconstruction. Biofilm-related maps and flagellar assembly were also detected. These findings suggest that the annotated gene repertoire contains functions that can participate in cell-cell communication, surface-associated growth, motility-related processes, and multicellular bacterial

behavior.

KEGG map	Pathway	Matched KO objects
02024	Quorum sensing	29
04112	Cell cycle - Caulobacter	9
05111	Biofilm formation - Vibrio cholerae	7
02026	Biofilm formation - Escherichia coli	3
02040	Flagellar assembly	2

S. aureus Infection and Antimicrobial-Resistance Pathways

Several biologically relevant maps occurred within the KEGG human-disease section. The Staphylococcus aureus infection pathway contained 12 matched KO objects. Antimicrobial-resistance maps included cationic antimicrobial peptide (CAMP) resistance with 10 matched objects, beta-lactam resistance with 9, vancomycin resistance with 5, and antifolate resistance with 4.

KEGG map	Pathway	Matched KO objects
01503	Cationic antimicrobial peptide (CAMP) resistance	10
01501	beta-Lactam resistance	9
01502	Vancomycin resistance	5
01523	Antifolate resistance	4

These pathway matches provide useful functional context for the resistome analysis because they identify KO-level functions participating in resistance-associated pathways. Importantly, a KEGG pathway match should not by itself be interpreted as proof of phenotypic resistance to an antimicrobial. Resistance conclusions should be integrated with gene-level resistance determinants, sequence context, and—where available—phenotypic susceptibility data.

Integrated COG–KEGG Interpretation

COG and KEGG provide complementary levels of functional interpretation. The COG profile describes the broad functional classes represented in the core gene repertoire, whereas KEGG links individual orthologous functions to metabolic pathways, information-processing systems, transport systems, modules, and higher-level pathway maps. The agreement between the two approaches strengthens the interpretation of a metabolically active and physiologically versatile S. aureus genome repertoire.

The high representation of COG categories E, C, G, F, and H is consistent with the KEGG enrichment of carbon metabolism, amino-acid biosynthesis, nucleotide metabolism, pyruvate metabolism, and cofactor biosynthesis. Likewise, the substantial COG representation of J, K, and L corresponds well with KEGG

ribosome, aminoacyl-tRNA biosynthesis, DNA replication, homologous recombination, and mismatch-repair pathways.

The COG M category for cell wall/membrane/envelope biogenesis is also consistent with the KEGG reconstruction of peptidoglycan biosynthesis and transport/secretion systems. The presence of COG defense functions and KEGG resistance pathways provides an additional functional layer that can be integrated with antimicrobial-resistance gene screening performed elsewhere in the project.

Important Interpretation and Quality Considerations

The KEGG reconstruction includes pathway maps from multiple biological contexts, including bacterial, plant, fungal, and animal systems. This occurs because homologous KO functions can be shared across organisms. Therefore, maps such as plant hormone signaling, MAPK signaling in eukaryotes, or human disease pathways should not be interpreted literally as evidence that S. aureus possesses the complete corresponding eukaryotic pathway. They are better treated as evidence that one or more conserved KO functions occur in the input dataset.

Similarly, the number shown beside an individual KEGG pathway represents matched objects in that pathway map, not necessarily the number of unique genes or a measure of pathway completeness. Many KO functions participate in multiple pathways, so pathway-level counts are not additive.

The COG result is percentage-based and therefore does not provide absolute gene counts. The high proportion in category S (function unknown; 21.34%) also indicates that a meaningful part of the core-gene repertoire remains poorly characterized at the COG functional level. This is a normal feature of genome annotation and represents an opportunity for additional domain, homology, structure, or literature-supported functional investigation.

Conclusion

The combined COG and KEGG analyses provide a comprehensive overview of the functional architecture of the analyzed S. aureus genome dataset. The COG profile demonstrates broad representation of metabolic, translational, transcriptional, ion-homeostasis, energy-conversion, carbohydrate-utilization, cell-envelope, and genome-maintenance functions. KEGG reconstruction extends this view by placing the annotated functions into 212 pathway entries and 37 modules, with particularly strong representation of central metabolism, biosynthetic pathways, ribosomal and DNA-maintenance systems, transporters, two-component signaling, quorum sensing, and cell-envelope processes.

Of particular project relevance, KEGG identified pathway-level matches for S. aureus infection and several antimicrobial-resistance mechanisms, including CAMP, beta-lactam, vancomycin, and antifolate resistance. These results establish a strong functional foundation for integrating pathway biology with the project's resistome, virulence, MLST, plasmid, and comparative-genomics analyses. Overall, the annotation profile supports the interpretation of S. aureus as a metabolically adaptable organism with extensive regulatory, transport, cell-envelope, stress-response, and host-associated functional potential.